Stephen Gavin Sears

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Profile:

Master's student in Computer Graphics and Game Technology at the University of Pennsylvania. Graduated as a Computer Science Major and Visual Studies Minor at Haverford College. Experienced in theoretical machine learning concepts and building custom ML pipelines. Tested coding and software design skills in C++, Python, SQL, C#, Swift, Java, JavaScript, and HTML/CSS, manually and with AI tools and agent workflows. Strong communicator with academic experience in computer graphics, machine learning, data science, data structures and algorithms. Work experience in web app development, machine learning, data analysis, IT, and AI agent workflows. Avid enthusiast of Git and GitHub, and proper repo management.

Education:

MSE Computer Graphics and Game Technology – University of Pennsylvania, Philadelphia, PA — Aug. 2024-Present
Expected Graduation Date: May 2026
Relevant Coursework: User Interface (UI) and User experience (UX) Design, Interactive Graphics, Procedural Computer Graphics, Computer Animation, Applied Machine Learning, Modern Convex Optimization, Advanced Topics in Computer Graphics

Computer Science Major & Visual Studies Minor - Haverford College, Haverford, PA — Aug. 2021-May 2025
Graduation Date: May 2025
Relevant Coursework: Machine Learning, Computer Graphics, Game Design, Game Programming, Analysis of Algorithms, Data Science, Data Structures, Discrete Mathematics, Computer Organization, Encoding Music, Multivariable Calculus, Financial Accounting

Work Experience:

Incoming Software Intern - Astronics Test Systems (40 hrs/week), Orlando, FL — June. 2026-Aug. 2026

- Architecting a voice-operated UI for hardware spectrum analyzers using Model Context Protocol (MCP) and Python/C# to enable real-time conversational control
- Developing front-end interfaces for AI-driven voice tools using HTML/JS/CSS, focusing on intuitive user experience for complex technical workflows
- Optimizing development with agentic AI workflows while maintaining strict code quality
- Managing secure data pipelines and machine learning models for confidential projects, ensuring compliance with rigorous aerospace security and architectural standards

Data Analyst and Power BI Specialist - Sears Mfg. Co. (40 hrs/week), Davenport, IA — May. 2025-Aug. 2025

- Created 6 fully automated financial and manufacturing Power BI reports that made data visible for managing supply chain and cost engineering
- Configured Data Lake views using Transact-SQL
- Generated Data Warehouse delta tables using scheduled pyspark notebooks and Fabric data pipelines
- Cleaned forecast data on request using python polars
- Transformed data for reports using Power Query, DAX

Teaching Assistant: Machine Learning – Haverford College, Haverford, PA — Jan. 2025-May 2025

- Held TA hours every week to assist machine learning students
- Graded student assignments on theoretical AI/ML concepts, Python, TensorFlow, and scikit-learn
- Assisted in testing and writing code for assignments

The HyperReal Theater - Bryn Mawr College, Bryn Mawr, PA — Aug. 2024-Dec.2025

- Met once a week to create VR and motion capture experiences in Prof. Aline Normoyle's lab
- Calibrated and prepared motion capture equipment for use in professional demos
- Tested and built game demos for the Oculus Quest
- Developed IK solvers in Unity C#, and rigged and textured 3D models for use with motion capture

Incubator Fellow - Haverford College (35 hrs/week), Haverford, PA — Jun. 2023-Jul. 2023

- Co-founded a game studio, Lemon Dragon, to produce Word Wizard, an English language learning video game, with a team of developers
- Scripted and designed dynamic UI systems for reading comprehension games using C# in Unity
- Managed and troubleshot file repositories hosted by GitHub using command line and GitHub desktop
- Utilized 3D modeling, texturing, and shading to create ready-to-use game assets
- Organized tasks for project using AGILE scrum framework

Programming Languages: C++, GLSL, Python, Java, JavaScript, C#, swift, Transact-SQL
Libraries and APIs: TensorFlow, pytorch, Polars, OpenGL, WebGL, Scene Kit, Unity C#, Pygame, Networkx, Pandas, Matplotlib, jQuery
Software: Google Antigravity, Claude code, codex, Qt, Visual Studio, Unity, VS code, Xcode, Microsoft Office, Excel, SQL Server Management Studio (SSMS), Lovable AI, Genspark